

Sol. (i) $\frac{3x}{2y} = \frac{21}{22} \Rightarrow \frac{x}{y} = \frac{21}{22} \times \frac{2}{3} = \frac{7}{11} \Rightarrow x = \frac{7}{11}y.$

$$4x + 5y = 83 \Rightarrow 4 \times \frac{7}{11}y + 5y = 83 \Rightarrow \frac{28}{11}y + 5y = 83$$

$$\Rightarrow \frac{83}{11}y = 83 \Rightarrow y = 83 \times \frac{11}{83} = 11.$$

$$\therefore x = \left(\frac{7}{11} \times 11 \right) = 7. \text{ So, } y - x = 11 - 7 = 4.$$

(ii) $\frac{x}{4} - \frac{x-3}{6} = 1 \Leftrightarrow \frac{3x - 2(x-3)}{12} = 1 \Leftrightarrow 3x - 2x + 6 = 12 \Leftrightarrow x = 6.$

Ex. 19. If $2x + 3y = 34$ and $\frac{x+y}{y} = \frac{13}{8}$, then find the value of $5y + 7x$.

Sol. The given equations are:

$$2x + 3y = 34 \quad \dots(i) \text{ and, } \frac{x+y}{y} = \frac{13}{8} \Rightarrow 8x + 8y = 13y \Rightarrow 8x - 5y = 0 \quad \dots(ii)$$

Multiplying (i) by 5, (ii) by 3 and adding, we get: $34x = 170$ or $x = 5$.

Putting $x = 5$ in (i), we get: $y = 8$.

$$\therefore 5y + 7x = (5 \times 8 + 7 \times 5) = 40 + 35 = 75.$$

Ex. 20. The cost of 4 bags and 3 boxes is ₹ 555 and the cost of 3 bags and 4 boxes is ₹ 460. What is the cost of one bag? (Bank P.O., 2009)

Sol. Let the cost of 1 bag be ₹ x and that of 1 box be ₹ y .

$$\text{Then, } 4x + 3y = 555 \quad \dots(i) \text{ and } 3x + 4y = 460 \quad \dots(ii)$$

$$\text{Adding (i) and (ii), we get: } 7x + 7y = 1015 \text{ or } x + y = 145 \quad \dots(iii)$$

$$\text{Subtracting (ii) from (i), we get: } x - y = 95 \quad \dots(iv)$$

$$\text{Adding (iii) and (iv), we get: } 2x = 240 \text{ or } x = 120.$$

Hence, cost of 1 bag = ₹ 120.

Ex. 21. If $2x + 3y + z = 55$, $x + z - y = 4$ and $y - x + z = 12$, then what are the values of x , y and z ?

Sol. The given equations are:

$$2x + 3y + z = 55 \quad \dots(i); \quad x + z - y = 4 \quad \dots(ii); \quad y - x + z = 12 \quad \dots(iii)$$

$$\text{Subtracting (ii) from (i), we get: } x + 4y = 51 \quad \dots(iv)$$

$$\text{Subtracting (iii) from (i), we get: } 3x + 2y = 43 \quad \dots(v)$$

$$\text{Multiplying (v) by 2 and subtracting (iv) from it, we get: } 5x = 35 \text{ or } x = 7.$$

$$\text{Putting } x = 7 \text{ in (iv), we get: } 4y = 44 \text{ or } y = 11.$$

$$\text{Putting } x = 7, y = 11 \text{ in (i), we get: } z = 8.$$

Ex. 22. If $x^2 - 7x = -12$, what is the value of x ? (M.B.A., 2006)

Sol. $x^2 - 7x = -12 \Leftrightarrow x^2 - 7x + 12 = 0 \Leftrightarrow x^2 - 3x - 4x + 12 = 0$

$$\Leftrightarrow x(x-3) - 4(x-3) = 0 \Leftrightarrow (x-3)(x-4) = 0 \Leftrightarrow x = 3 \text{ or } x = 4.$$

Ex. 23. Find the value of $\left(1 - \frac{1}{3}\right)\left(1 - \frac{1}{4}\right)\left(1 - \frac{1}{5}\right) \dots \left(1 - \frac{1}{100}\right)$.

Sol. Given expression $= \frac{2}{3} \times \frac{3}{4} \times \frac{4}{5} \times \dots \times \frac{99}{100} = \frac{2}{100} = \frac{1}{50}.$

Ex. 24. Find the value of $\frac{1}{2 \times 3} + \frac{1}{3 \times 4} + \frac{1}{4 \times 5} + \frac{1}{5 \times 6} + \dots + \frac{1}{9 \times 10}.$

Sol. Given expression $= \left(\frac{1}{2} - \frac{1}{3}\right) + \left(\frac{1}{3} - \frac{1}{4}\right) + \left(\frac{1}{4} - \frac{1}{5}\right) + \left(\frac{1}{5} - \frac{1}{6}\right) + \dots + \left(\frac{1}{9} - \frac{1}{10}\right)$
 $= \left(\frac{1}{2} - \frac{1}{10}\right) = \frac{4}{10} = \frac{2}{5}.$